Problem 26.14

A 20 year annuity-immediate pays $500 + 50t$ at the end of year t. Calculate the present value of this annuity using an annual effective rate of 6%.

Solution

At the end of year t, the payment is

$$500 + 50t.$$

Therefore, the payment stream can be separated into a level annuity of 500 and an increasing annuity with payments $50, 100, 150, \ldots, 1000$. Thus,

$$PV = 500a_{\overline{20}|} + 50(Ia)_{\overline{20}|}.$$

At $i = 0.06$,

$$v = \frac{1}{1.06}$$

and

$$a_{\overline{20}|} = \frac{1 - v^{20}}{0.06} = 11.46992122.$$

For an increasing annuity-immediate,

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}.$$

Since

$$\ddot{a}_{\overline{20}|} = (1.06)a_{\overline{20}|} = 12.15811649,$$

we have

$$(Ia)_{\overline{20}|} = \frac{12.15811649 - 20(1.06)^{-20}}{0.06} = 98.70036590.$$

Therefore,

$$PV = 500(11.46992122) + 50(98.70036590).$$

Thus,

$$PV = 10669.97890.$$

Hence, the present value is

$$\boxed{10669.98}.$$